



The Wonders of the Waterfowl World

Description:

Students discover how form and function are linked in the world of ducks and geese. Through the observation of waterfowl characteristics and other avian variations, students will understand how certain attributes provide the waterfowl species with advantages and/or disadvantages for life on Core Sound.

Goal:

The learner will make observations and conduct investigations to build an understanding of animal behavior and adaptation.

Objectives:

Students will be able to:

- (1) Observe and record how waterfowl differ in some of their characteristics from other birds
- (2) Discuss possible advantages and disadvantages of variation
- (3) Explain how certain behaviors, body structures, and plumage help waterfowl survive in Core Sound
- (4) Draw an accurate representation of a waterfowl based on knowledge gained

Grade Level: 4th

Subject Area: Science

NCSCOS Objective(s):

Science: 1.02-1.03

Duration: 1 hour

Group Size: 25

Setting: Indoor/Outdoor
at Willow Pond Trail

Key Vocabulary:

- Waterfowl
- Variation
- Plumage
- Decoy
- Drake
- Hen

Materials:

- Paper
- Pencil, Markers, Crayons
- White board or large sheet of paper
- Power Point "The Wonders of the Waterfowl World"
 - Laptop, Projector, Screen
- Worksheet "The Wonders of the Waterfowl World"
- Color/Paint samples
- Photos/Decoys of different species of waterfowl
- List of the 23 Functions of Feathers



Background Information:

Waterfowl are a family of birds that include the ducks, geese and swans. The scientific name for this family is Anatidae (ah NAT ih day). There are almost 50 different species of waterfowl in North America, 44 species have been identified in North Carolina, and there are approximately 150 species worldwide.

General waterfowl characteristics:

- Body: relatively long with a rounded shape
- Feathers: coated with a waterproof oil
- Wings: pointed and built for flying long distances
- Neck: medium to long neck, fairly slim, stretches forward during flight
- Bill: wide and flat with tooth-like edges, hard "nail" at the tip
- Tail: short compared to the rest of their body
- Legs: set back on the body to aid in swimming (causes them to waddle on land)
- Toes: 4 on each foot, front 3 toes are webbed and used for paddling

Ducks make up the largest group of the waterfowl family. They are typically smaller in size than the geese and swans. Plumage varies between males (drakes) and females (hens). The drakes have brightly colored plumage (feathers) with bold patterns (especially during breeding season) to attract the females. The drake wood duck is one of the most colorful. The hens are often dull-colored or brown in order to camouflage themselves and their young from predators. Ducks live in lakes, rivers, freshwater wetlands, salt marshes, and the ocean. Their diet is species dependent and includes plants and/or small animals. Some ducks have specialized bills that are designed to help them gather food. For example, the northern shoveler has a bill that is wide and flat with comb-like edges for straining food from the water. The common merganser's bill is long and thin with extra-sharp edges for catching fish.

Ducks are divided into two different groups, the dabblers and the divers, based on how and where they feed. Dabblers are usually found in shallow water. They "tip up" and dunk their head underwater to feed. Dabblers tend to sit high on the water, have plump, round bodies, and can walk pretty well on land. They also have long, wide wings that allow them to take off straight into the air. Divers tend to spend more time in deep water. They have the ability to dive completely underwater in search of food. Divers sit lower in the water. They have barrel-shaped bodies and shorter necks. Their legs are set farther back on their bodies to help them dive, which also means that they have an even more difficult time walking. Divers have shorter, narrower wings which assist with swimming, but not with taking off. They have to run across the water to gain enough speed to take off.

Geese are usually bigger than ducks and smaller than swans. They also have less colorful feathers than many of the male ducks. Geese are comfortable in the water and on land. They are the nosiest and most social members of the waterfowl family, nesting in large groups called colonies. Goose families flock together when they migrate in the spring and fall. Male and female geese look alike, but the male (gander) is usually larger than the female. Males and females pair for life and raise their goslings together. All geese are plant eaters. Their bills have sharp edges for clipping and tearing leaves and grass.

The Canada goose is the largest of all the geese, weighing as much as 20 pounds. It is also the most common goose found in North America, which is home to 8 different species of geese.

Swans are the largest members of the waterfowl family. Their necks are actually longer than their bodies, and because of their size, they must run across the water's surface to reach flight speed. All of North America's 3 species of swans are white. They eat leaves and grass and use their long necks to pull up aquatic plants. The male and female swans also pair for life and raise their young together.

The trumpeter swan is the largest species of waterfowl. Some weigh up to 30 pounds and have wings 8 feet across. These birds are endangered species.

Other birds that are often mistaken for waterfowl are coots, grebes, loons, and gulls; however they do not share the same general characteristics of ducks, geese, and swans.

- Coots: chicken-like beak, no webbed toes
- Grebes: pointed beaks, no webbed toes
- Loons: large, heavy bill
- Gulls: hooked beaks and wings shaped differently than waterfowl wings

Procedure:

Warm-Up:

1. Ask students to close their eyes and image a duck. Then tell them to open their eyes and draw what they saw in their minds with a pencil on a sheet of blank paper.
2. Have the students share their drawings with at least 2 other students.
3. As a group, create a duck concept map that reflects what the students drew.

Activity:

1. Engage the students the interactive PowerPoint presentation "Wonders of the Waterfowl World."
The presentation:
 - a. provides general information on birds/waterfowl
 - b. highlights the variation within the avian world
 - c. asks students to identify key characteristics of waterfowl
 - d. discusses the advantages of particular characteristics
2. Take students outside to view an outdoor landscape (Willow Pond, Core Sound, salt marsh, maritime forest, etc.) around the museum. Discuss color in nature. Trees are not just green; water is more than one shade of blue, etc.

3. Have students select one or several color chips and match them with something in the landscape they are viewing.
4. Ask students to describe the color differently than the name already assigned to the color chip or use similes, metaphors, or other descriptors to describe the color.
5. Back in the museum, have students match their color chips with feather colors on decoys or waterfowl photographs.
6. Discuss the ways that birds use feathers and talk about the advantages/disadvantages of bird coloration/plumage.
 - a. Optional: Use the 23 Functions of Feathers list. Cut the list into strips and have students draw the functions out of a bowl. Ask them to read the function aloud or draw a picture that represents the function to share with the group.

Wrap-Up:

7. Ask students to develop a new drawing of a duck (waterfowl) or change the existing one to correspond with the new information they have learned. Ask them to take into account variation in bird characteristics and the environments of Core Sound. Provide crayons (color).
8. Have the students compare their before and after drawings with a partner and discuss the reasons for the changes made.
9. Have students create a new concept map or change the existing one to correspond to the waterfowl of Core Sound.

Assessment:

Have students:

- Name types of waterfowl
- Describe key characteristics of waterfowl
- Create a drawing that accurately represents a duck
- Assemble a concept map of waterfowl characteristics that provide an advantage to the ducks, geese, and swans of Core Sound

Extensions:

Have students paint watercolors, decoy heads, or decoys that replicate the plumage of their favorite waterfowl.

Resources:

Brakhage, David. "Waterfowl Feathers." Ducks Unlimited Waterfowl Biology. Ducks Unlimited, n.d. Web. 29 Jul 2011.
<http://www.ducks.org/conservation/waterfowl-biology/waterfowl-feathers>.

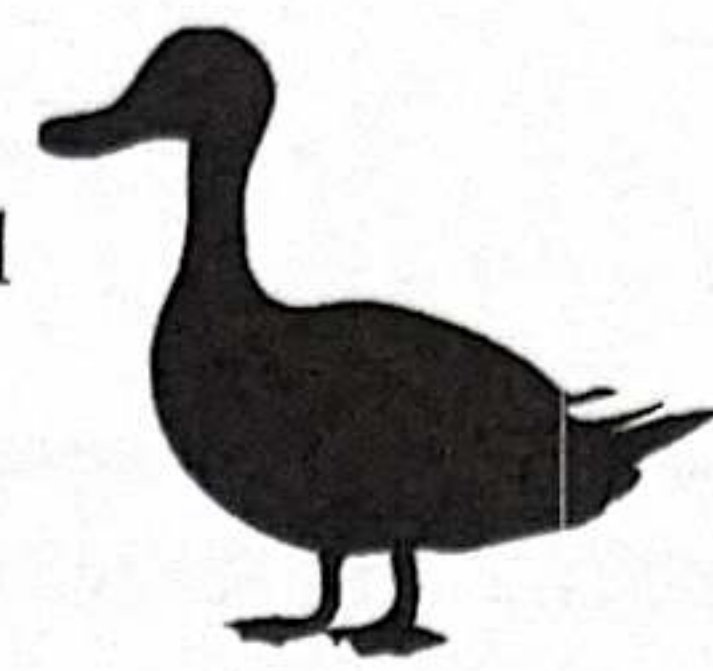
"Feather Biology." Ask a Biologist. Arizona State University, n.d. Web. 29 July 2011.
<http://askabiologist.asu.edu/content/23-functions-feathers>.

Jones, David. *Ducks*. Vancouver/Toronto, Canada: Whitecap Books, 1998. Print.

Koch, Maryjo. *Bird Egg Feather Nest*. San Francisco, CA: Collins Publishers, 1994. Print.

"The Waterfowl Family." *Puddler*. Fall 2008: 4-9. Print.

The Wonders of the Waterfowl World



Name: _____

1. What groups of animals do waterfowl belong to?

a. Fish

b. Reptiles

c. Birds

d. Mammals

2. Why is a bird's skeleton strong and lightweight? _____

3. Why are feathers important for a bird? _____

4. List one characteristic that you share with birds: _____

5. Which flyway do birds in North Carolina use for migration?

a. Atlantic Flyway

b. Mississippi Flyway

c. Central Flyway

d. Pacific Flyway

6. The waterfowl family is made up of which three types of birds? Circle all that apply:

a. Parrots

b. Ducks

c. Penguins

d. Geese

e. Swans

f. Chickens

7. Why do ducks have wide, flat bills? _____

8. Which feature makes ducks good swimmers?

a. Wings

b. Feathers

c. Skeleton

d. Toes

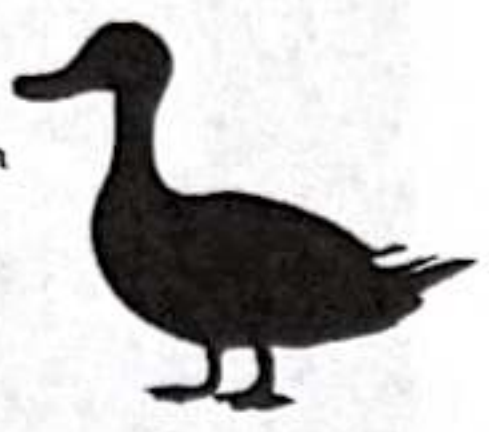
9. A male duck is called a _____. A female duck is called a _____.

10. Why do some waterfowl have brightly colored plumage/feathers? _____

11. Why do some waterfowl have brown or dull colored plumage/feathers? _____

THE WONDERS OF THE WATERFOWL WORLD

Core Sound
Waterfowl Museum
&
Heritage Center



Waterfowl are Birds



Kingdom: Animalia Phylum: Chordata
Class: Aves

**10,000
Living Species of Birds Worldwide**

POLAR PENGUINS TROPICAL PARROTS




**Birds
Come in Many Shapes and Sizes:**



2 INCH
BEE HUMMINGBIRD



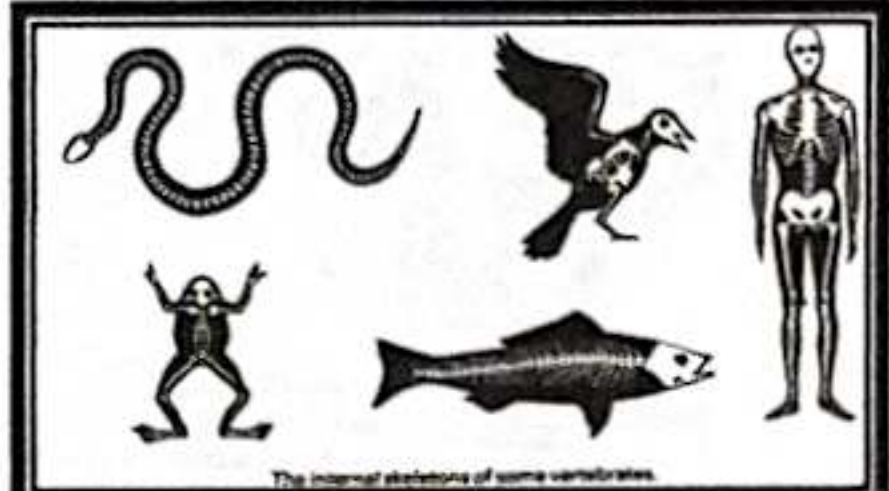
9 FOOT
OSTRICH

**CHARACTERISTICS OF
ALL BIRDS**

Including Waterfowl...

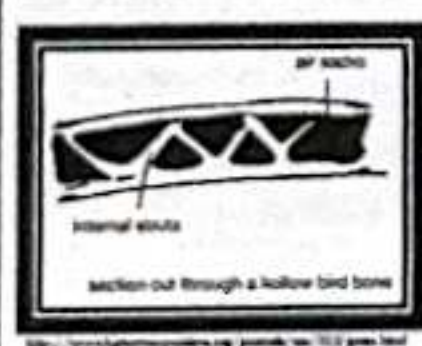


**Vertebrates:
Have A Backbone**



The internal skeletons of some vertebrates.

**Skeleton:
Strong & Lightweight**

section cut through a hollow bird bone

**Endothermic:
Generate Body Heat**



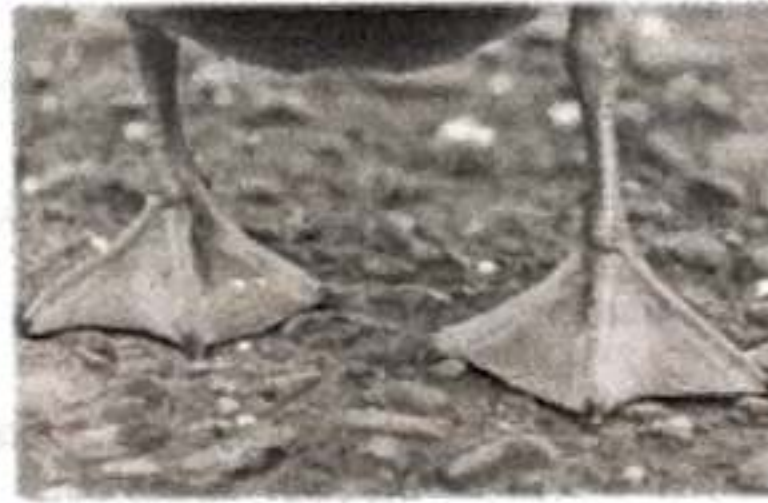
**Feathers:
For Flight, Warmth & Coloration**



**Wings:
For Flight**



**Bipedal:
Walk On Two Legs/Feet**



**Beak:
Specialized for Feeding**



Lay Eggs



**Social:
Flock Together & Communicate**



Many Migrate



**The Waterfowl Family:
Ducks, Geese & Swans**

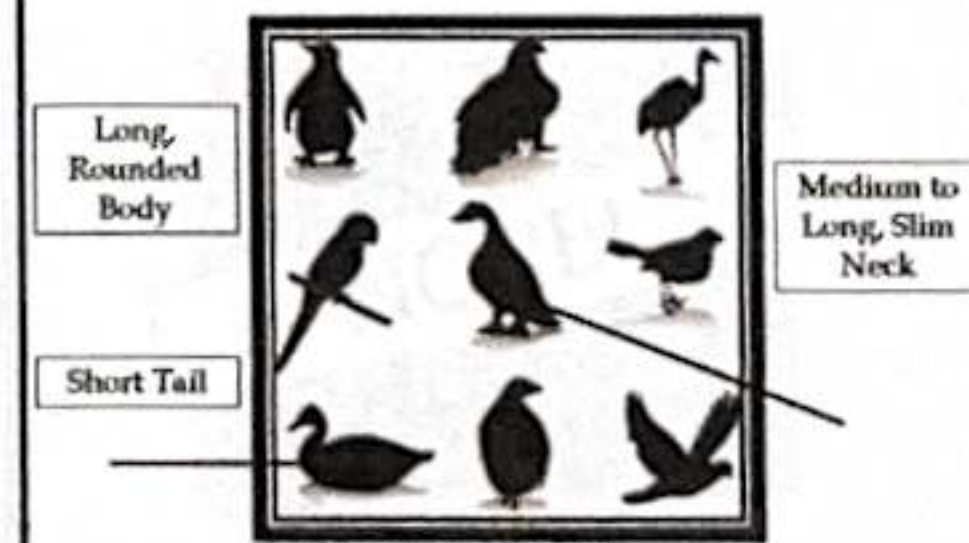


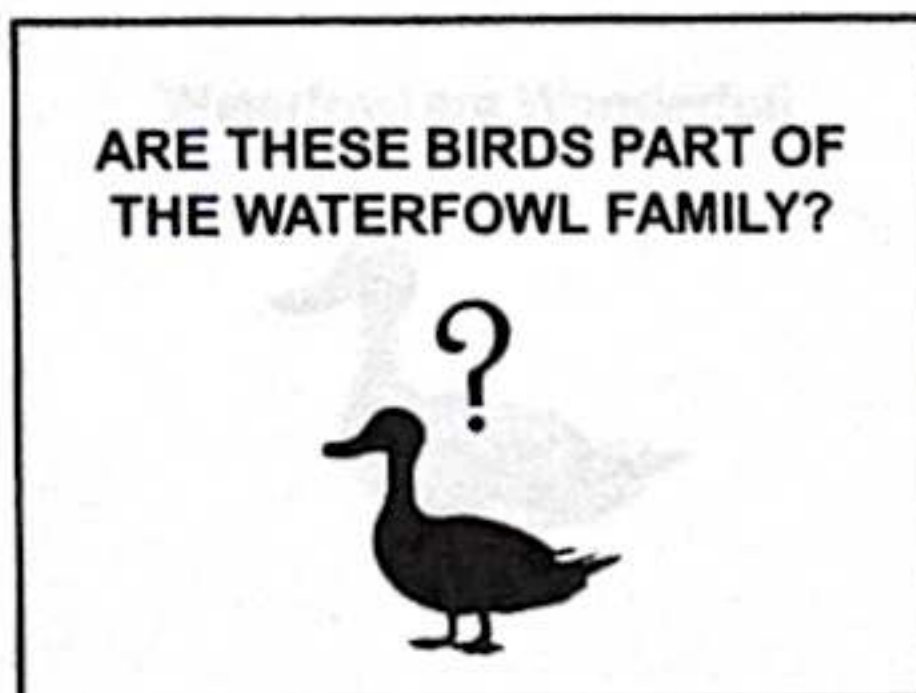
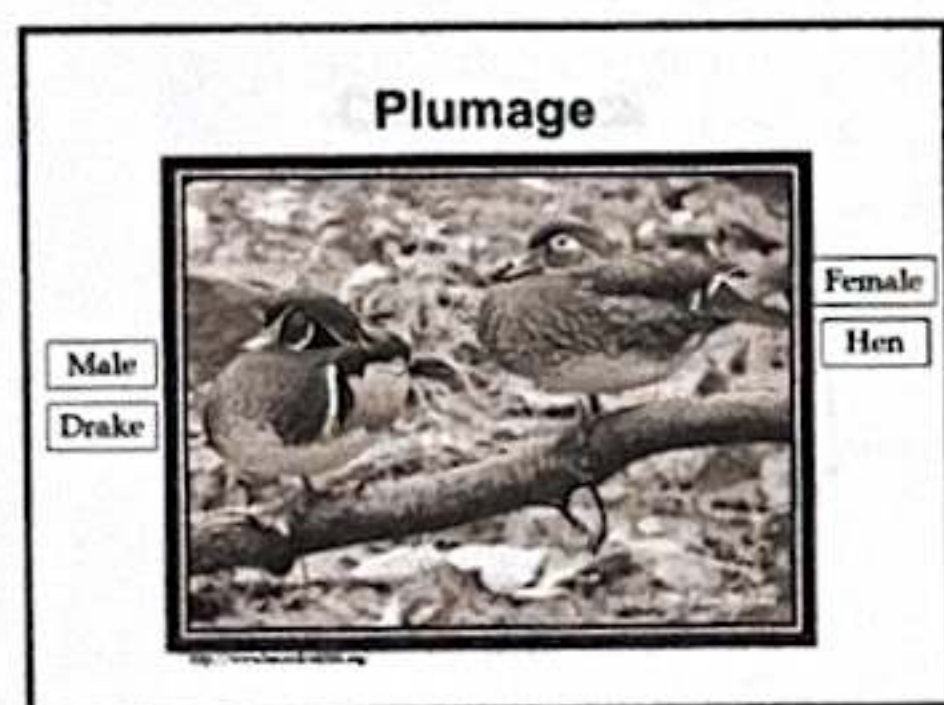
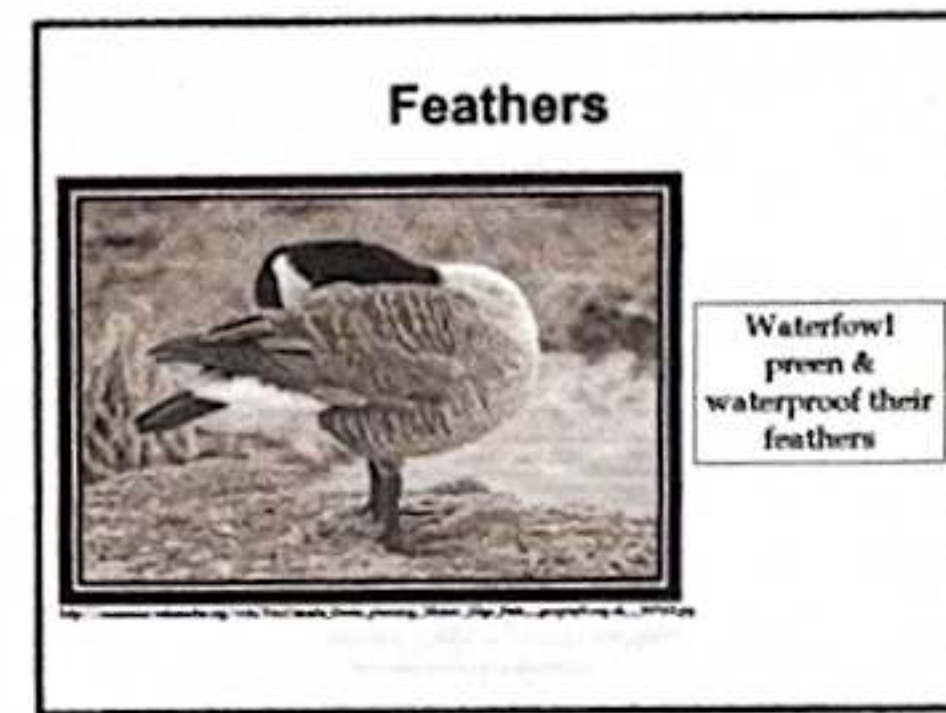
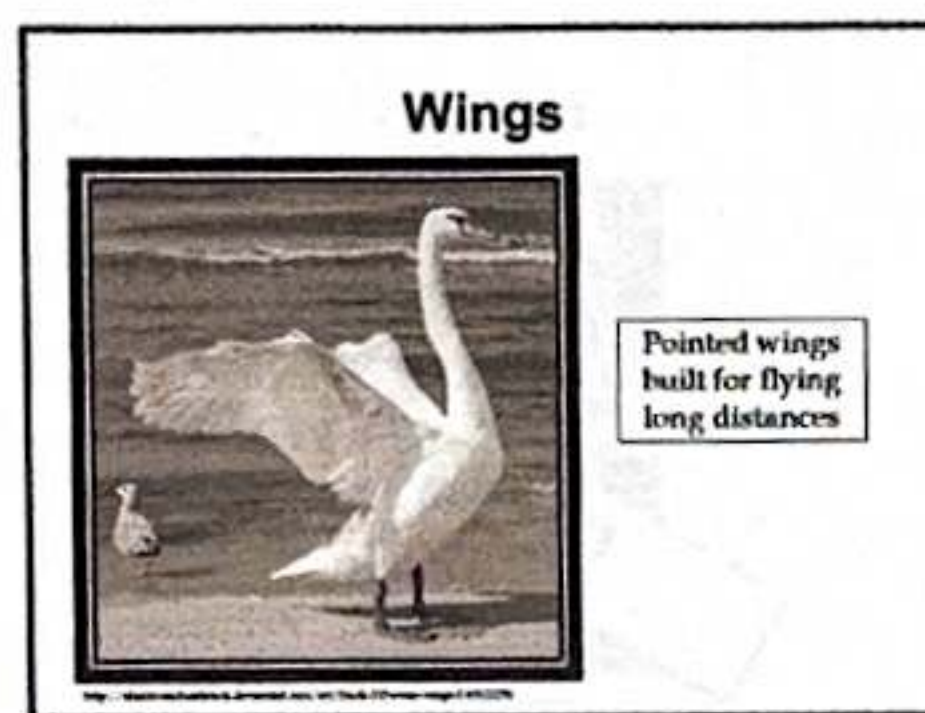
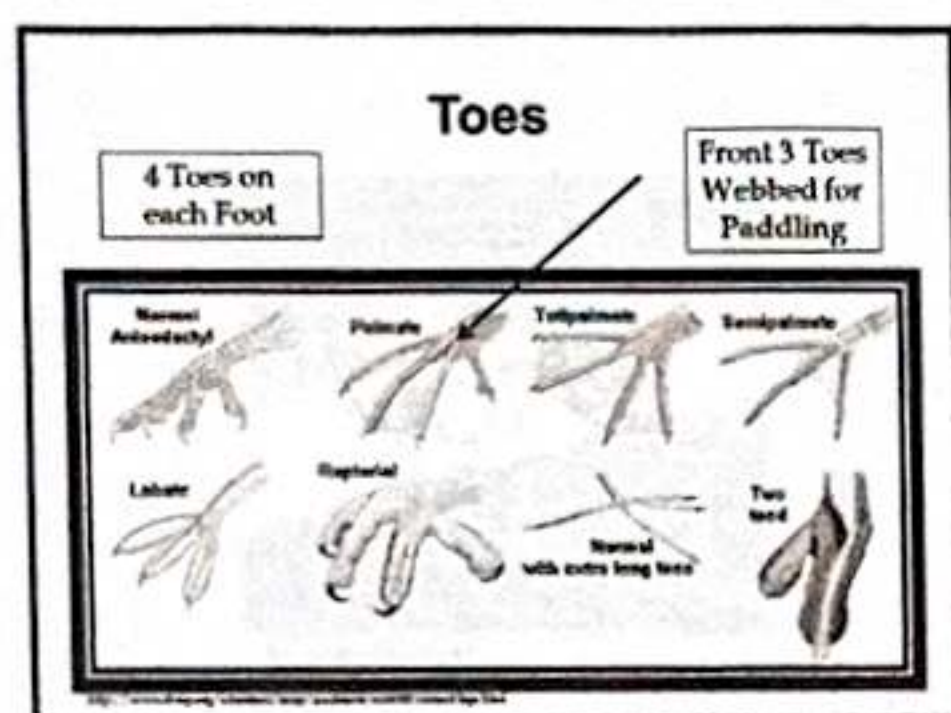
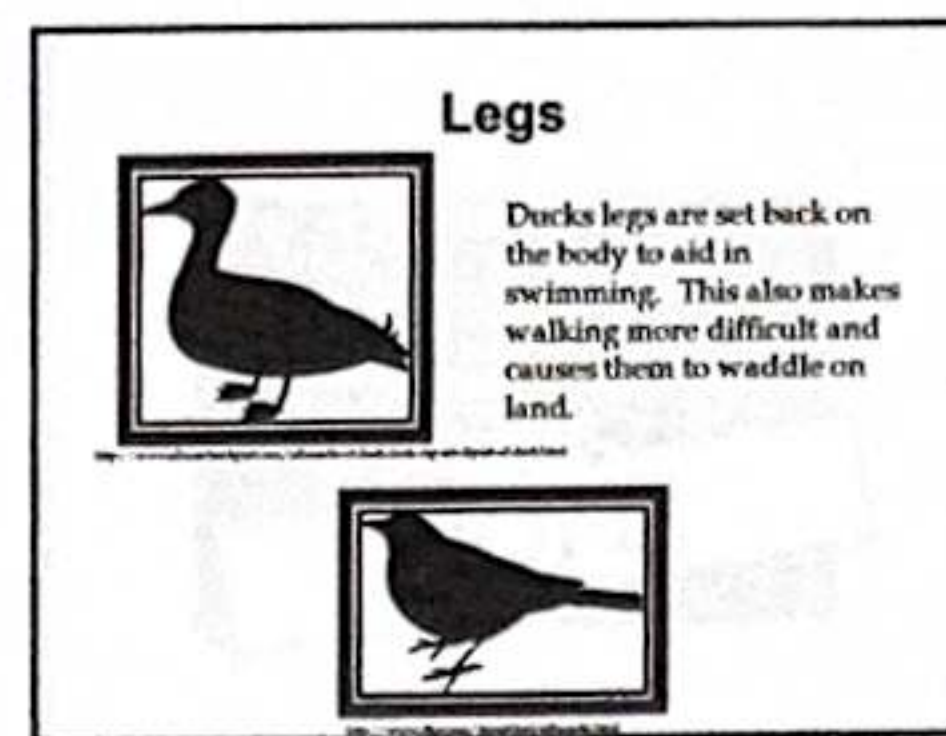
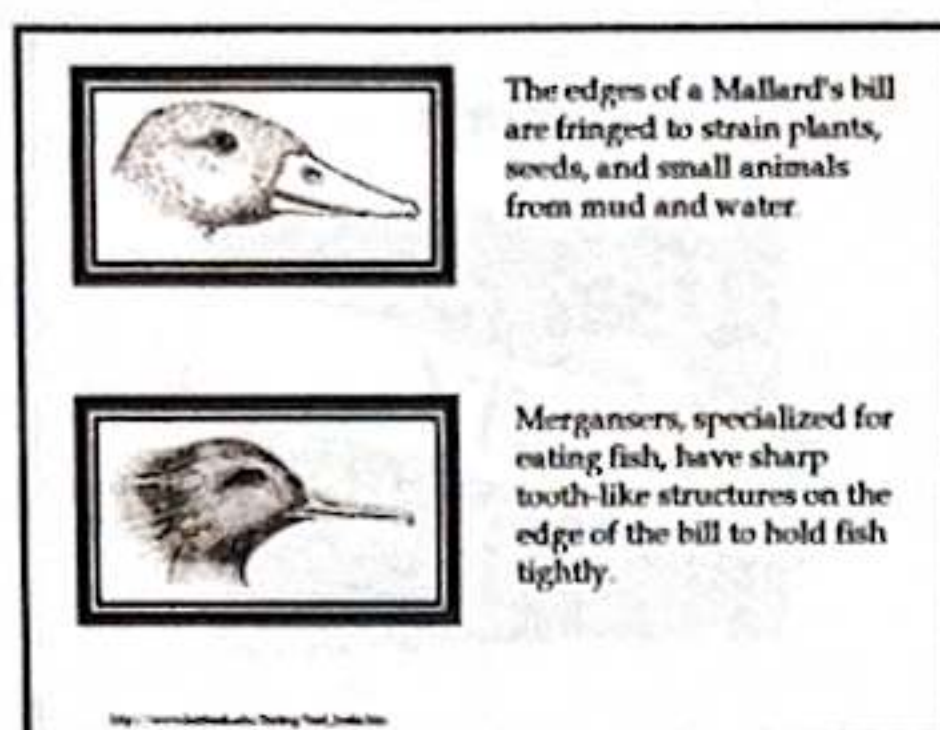
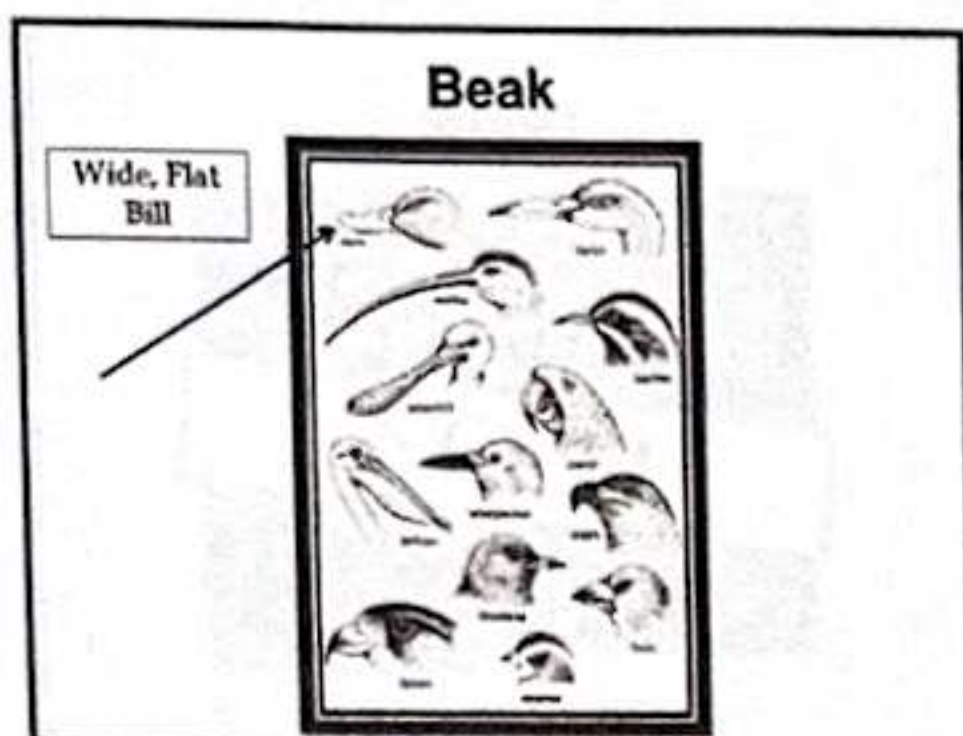
There are 44 different species of waterfowl in North Carolina.

**WHAT MAKES THE
WATERFOWL FAMILY
SPECIAL?**



Body Shape





Grebe



Pintail



Loon



Gull



Pelican



Heron



Canvasback



Waterfowl are Wonderful!

